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Transmission type optical measurement device having improved measurement accuracy

Abstract

A conveyance device supports and conveys an object. The conveyance device has a support portion in which an opening narrower than the object is provided at a position where the object is supported. A lighting device irradiates a first surface of the object with measurement light having a wavelength changing over time through the opening of the support portion. A light receiving device detects object light that is diffusely transmitted light emitted from a second surface of the object.

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Background/Summary

BACKGROUND
1. Technical Field
(1) The present disclosure relates to an optical measurement device.
2. Description of the Related Art
(2) Spectroscopic analysis is widely used for component analysis and inspection of an object. In the spectroscopic analysis, the object is irradiated with measurement light, and a spectrum of object light obtained as a result of the irradiation is measured. Then, optical characteristics such as reflection characteristics (wavelength dependency) or transmission characteristics can be obtained based on a relationship between the spectrum of the object light and the spectrum of the measurement light.
(3) Spectroscopic analysis is classified into a transmission type in which transmitted light of the object is the object light and a reflection type in which reflected light is the object light. The reflection type is suitable for measuring an object having a high reflectance, but the obtained optical information is limited to the optical information near the surface of the object. Therefore, it

cannot be said that the measurement has sufficient accuracy in a measurement of a precise industrial product, a specimen collected from an animal and a plant, a material taken into a human body, a liquid or gas produced in a production plant, or the like as the object.
(4) The transmission type can obtain optical characteristics including not only the surface but also a deep portion of the object, and thus is suitable for a case where a food, a beverage (hereinafter, collectively referred to as food and drink.), or the like is used as the object. Patent Literature 1 (JP2020-159971A) and Patent Literature 2 (JP2020-159973A) disclose product inspection devices of a transmission type. The product inspection devices each includes an irradiation optical system in which a front surface of the product (object to be inspected) is irradiated with pulsed light, and a light receiver provided on a back surface side of the product and configured to receive light transmitted through the product.
(5) The present inventor has studied an inspection device of the transmission type, and as a result, has recognized the following problems. FIG. **1** is a view illustrating an inspection device **1000** of the transmission type studied by the present inventor. Note that the inspection device **1000** should not be recognized as a known technique.
(6) A product P is supported and conveyed by a support means **1400**.
(7) As disclosed in Patent Literatures 1 and 2, a lighting device **1200** irradiates an upper surface of the product P with pulsed light (measurement light S.sub.IN). A light receiver **1300** is provided below the product P and receives object light S.sub.OBJ emitted from a bottom surface side of the product P. The object light S.sub.OBJ contains diffusely transmitted light as a main component, and is emitted from the entire bottom surface of the product P.
(8) When the support means **1400** is made of a material transparent to the object light S.sub.OBJ, the cost increases, and thus the support means **1400** opaque to the object light S.sub.OBJ may be used. In this case, an opening **1102** for taking out the object light S.sub.OBJ is required immediately below the product P of the support means **1400**. When the opening **1102** is too large, the product P falls, and thus the size (width) of the opening **1102** is required to be narrower than the product P.
(9) Only the object light S.sub.OBJ having passed through the opening **1102** can be incident on the light receiver **1300**, and a part of the object light S.sub.OBJ of the product P is shielded by the support means **1400**. As the width of the opening **1102** is narrowed, the amount (intensity) of the object light S.sub.OBJ incident on the light receiver **1300** decreases, leading to a decrease in measurement accuracy.
(10) In order to increase the amount of the object light S.sub.OBJ incident on the light receiver **1300**, the intensity of the measurement light S.sub.IN may be increased. However, depending on the type of the product P, there is a problem that the irradiation with the strong measurement light S.sub.IN changes the quality of the product P. Therefore, the intensity of the measurement light S.sub.IN with which the product P can be irradiated may have an upper limit I.sub.MAX. The maximum incident intensity I.sub.DET(MAX) on the light receiver **1300** is expressed as
(11) $I_{DET(MAX)} = I_{MAX} \times \quad \times \quad$, where transmittance of the product P is α, and a shielding rate (transmittance) of the support means **1400** is β. Thus, the amount of received light is reduced by the support means **1400**. A decrease in the amount of received light leads to a decrease in measurement accuracy.
(12) When a material such as glass is used as the support means **1400**, the opening **1102** need not be required, but even in this case, the transmittance β cannot be 100%. Therefore, the maximum incident intensity I.sub.DET to the light receiver **1300** is I.sub.DET=I.sub.MAX×α×β, and the amount of received light is reduced by the support means **1400**. A decrease in the amount of received light leads to a decrease in measurement accuracy.
SUMMARY
(13) The present disclosure has been made in view of the above problems, and one of exemplary objects of a certain aspect thereof is to provide an optical measurement device with improved

measurement accuracy.
(14) A certain aspect of the present disclosure relates to an optical measurement device. The optical measurement device includes a conveyance device that supports and conveys an object and includes a support portion that supports the object, a lighting device that irradiates a first surface of the object with measurement light having a wavelength changing over time through the support portion, and a light receiving device that detects diffusely transmitted light emitted from a second surface of the object.
(15) Note that any combination of the above components and mutual replacement of the components and expressions of the present disclosure among methods, devices, systems, and the like are also effective as aspects of the present disclosure.

Description

BRIEF DESCRIPTION OF DRAWINGS
(1) Embodiments will now be described, by way of example only, with reference to the accompanying drawings which are meant to be exemplary, not limiting, and wherein like elements are numbered alike in several Figures, in which:
(2) FIG. **1** is a view illustrating an inspection device of a transmission type studied by the present inventor.
(3) FIG. **2** is a block diagram of an optical measurement device according to an embodiment.
(4) FIG. **3** is a graph showing measurement light S.sub.IN.
(5) FIG. **4** is a graph for explaining spectroscopy by the optical measurement device in FIG. **2**.
(6) FIG. **5** is a view illustrating an optical measurement device according to Example 1.
(7) FIG. **6** is a view illustrating an optical measurement device according to Example 2.
(8) FIG. **7** is a view illustrating an optical measurement device according to Example 3.
(9) FIGS. **8**A to **8**C are views each illustrating a configuration example of a support portion **401**.
(10) FIG. **9** is a view illustrating an optical measurement device according to Example 5.
(11) FIG. **10** is a view illustrating an optical measurement device according to Example 7.
(12) FIG. **11** is a view illustrating an optical measurement device according to a comparative technique.
(13) FIG. **12** is a view illustrating a configuration example of a light receiving device.
(14) FIG. **13** is a view illustrating an inspection device that is one embodiment of an optical measurement device.
DESCRIPTION OF EMBODIMENTS
Outline of Embodiments
(15) An outline of some exemplary embodiments of the present disclosure will be described. This outline describes some concepts of one or more embodiments in a simplified manner for the purpose of basic understanding of the embodiments as a prelude to the detailed description that follows, and is not intended to limit the breadth of the invention or disclosure. Also, this outline is not a comprehensive outline of all possible embodiments and is not intended to limit the essential components of the embodiments. For convenience, the term “one embodiment” may be used to refer to one embodiment (example or modification) or a plurality of embodiments (examples or modifications) disclosed in the present specification.
(16) The optical measurement device according to one embodiment includes a conveyance device that supports and conveys an object and includes a support portion that supports the object, a lighting device that irradiates a first surface of the object with measurement light having a wavelength changing over time through the support portion, and a light receiving device that detects diffusely transmitted light emitted from a second surface of the object.
(17) According to this configuration, the diffusely transmitted light emitted from the object is not

shielded by the support portion, and more light can be detected by the light receiving device, so that the measurement accuracy can be improved.
(18) Note that the term “support” includes not only fixing the object but also holding the object in a certain range.
(19) In one embodiment, the support portion may have an opening narrower than the object. The lighting device may irradiate the first surface of the object with the measurement light through the opening of the support portion.
(20) In one embodiment, the conveyance device may include a plurality of support portions each supporting the object, and the plurality of support portions may each include a recessed portion provided on a common surface of the conveyance device.
(21) The opening may be a through hole formed in a bottom surface of the recessed portion.
(22) In one embodiment, the lighting device may include a folding mirror provided on a lower side of the conveyance device, and the folding mirror may reflect the measurement light incident from a side surface of the conveyance device towards the first face of the object.
(23) In one embodiment, the lighting device may be provided on the lower side of the conveyance device without using the folding mirror.
(24) In one embodiment, the light receiving device may include a light sensor having a size (outer dimension) smaller than the object, and a condensing optical system including a lens having a size (outer dimension) larger than the object. The present inventors have recognized that when a specific object is to be inspected, diffusely transmitted light that is the object light has low directivity and is emitted over a wide range. In this case, by using the lens larger than the object, the diffusely transmitted light can be condensed on the light sensor as much as possible.
(25) In one embodiment, the light receiving device may include a light sensor, and may be configured such that a component of the diffusely transmitted light of the object emitted in a direction deviated from an optical axis of the measurement light is incident on the light sensor.
(26) Note that the term “configured” is not limited to a case where there is a feature in the configuration, and includes a case where there is a feature in both the configuration and the arrangement, a case where there is a feature only in the arrangement, and the like. According to this optical measurement device, when the object is present, the object light attenuated by the object is incident on the light sensor, and when the object is not present, the measurement light is not incident on the light sensor, or even if the measurement light is incident, the intensity is very weak, so that the light sensor can be protected. In addition, the lighting device can be continuously operated regardless of presence or absence of the object, and a shutter or the like synchronized with the presence or absence of the object is not required.
(27) In one embodiment, the measurement light may have a wavelength changing over time. In one embodiment, the measurement light may be pulsed light having a wavelength changing over time within one pulse.
Embodiment
(28) Hereinafter, the present disclosure will be described based on a preferred embodiment with reference to the drawings. The same or equivalent components, members, and processing illustrated in the drawings are denoted by the same reference numerals, and redundant description will be omitted as appropriate. In addition, the exemplary embodiment is not intended to limit the disclosure, but is merely an example, and all features described in the exemplary embodiment and combinations thereof are not necessarily essential to the disclosure.
(29) Dimensions (thickness, length, width, and the like) of each member described in the drawings may be enlarged or reduced for easy understanding as appropriate. Furthermore, the dimensions of the plurality of members do not necessarily indicate the magnitude relationship therebetween, and even if a certain member A is drawn thicker than another member B in the drawing, the member A may be thinner than the member B.
(30) FIG. **2** is a block diagram of an optical measurement device **100** according to the embodiment.

The optical measurement device **100** is a spectrometer that measures a transmission spectrum of an object OBJ, and mainly includes a lighting device **200**, a light receiving device **300**, a conveyance device **400**, and a processing device **500**. In some drawings, the lighting device **200**, the light receiving device **300**, and the like may be illustrated in a simplified form as a box, but this is not intended to house the respective constituent members in a single housing.
(31) The conveyance device **400** conveys the object OBJ so as to cross an irradiation region **10**.
(32) The conveyance device **400** includes a support portion **401** and the object OBJ is supported in the support portion **401**. The support portion **401** has an opening **402** narrower than the object OBJ at a position where the object OBJ is supported, and the support portion **401** supports the object OBJ such that the object OBJ straddles the opening **402**. The width of the opening **402** is determined so that the object OBJ does not fall from the opening **402**, and thus the width of the opening **402** may be narrower than the length of the object OBJ in the width direction.
(33) The lighting device **200** irradiates the first surface (bottom surface) of the object OBJ present in the irradiation region **10** with the measurement light S.sub.IN having the wavelength changing over time from a lower side of the support portion **401** through the opening **402**. In order to prevent the measurement light S.sub.IN from being shielded by the support portion **401**, a beam diameter of the measurement light S.sub.IN is condensed smaller than the opening **402**.
(34) In the measurement light S.sub.IN, a time and a wavelength are associated with each other in a one-to-one relationship. This is referred to as the measurement light S.sub.IN “having wavelength uniqueness”. The lighting device **200** may be configured using a known technique, and for example, those described in Patent Literatures 1 and 2 can be used.
(35) FIG. **3** is a graph showing the measurement light S.sub.IN. The upper part of FIG. **3** shows the intensity (time waveform) I.sub.IN(t) of the measurement light S.sub.IN, and the lower part illustrates the time change of the wavelength λ of the measurement light S.sub.IN.
(36) In this example, the measurement light S.sub.IN is one pulse, the dominant wavelength is λ.sub.1 at the leading edge, the dominant wavelength is λ.sub.2 at the trailing edge, and the wavelength changes over time between λ.sub.1 and λ.sub.2 within one pulse. In this example, the measurement light S.sub.IN is a positive chirp pulse (λ.sub.1>λ.sub.2) having a frequency increasing with time, in other words, having a wavelength decreasing with time. Note that the measurement light S.sub.IN may be a negative chirp pulse having a wavelength increasing with time (λ.sub.1<λ.sub.2).
(37) Returning to FIG. **2**. The measurement light S.sub.IN is radiated to the bottom surface side of the object OBJ, passes through the object OBJ, and is emitted as transmitted light (Hereinafter, also referred to as object light.) S.sub.OBJ from the second surface (upper surface). The spectrum I.sub.OBJ(λ) of the object light S.sub.OBJ is expressed by an equation, where the spectrum of the measurement light S.sub.IN is I.sub.IN(λ) and the wavelength dependence of the transmittance of the object light S.sub.OBJ is T(λ).
(38) $I_{OBJ}(\ ) = T(\ ) \times I_{IN}(\ )$ (1)
(39) The object light S.sub.OBJ may include normally transmitted light and diffusely transmitted light, but the present embodiment is particularly suitable for spectroscopic measurement of an object OBJ in which the diffusely transmitted light is dominant. The normally transmitted light is emitted in the same direction as an optical axis OA**2** of the measurement light S.sub.IN, whereas the object light S.sub.OBJ that is the diffusely transmitted light is widely emitted not only in a direction of the optical axis OA**2** of the measurement light S.sub.IN but also in a direction different from the direction thereof. For example, when the direction of the optical axis OA**2** is set to 0°, the diffusely transmitted light is emitted with an intensity distribution of cosine characteristics.
(40) The light receiving device **300** is provided on the opposite side of the lighting device **200** across the support portion **401**, in other words, on an upper side of the support portion **401**, and detects the diffusely transmitted light emitted from the upper surface of the object OBJ. The light

receiving device **300** includes a light sensor **302** that detects the diffusely transmitted light of the object OBJ as the object light S.sub.OBJ. The light receiving device **300** may include a condensing optical system and the like in addition to the light sensor **302** as described later, but they are omitted in FIG. **2**.

(41) The light sensor **302** is a photoelectric conversion element that converts an optical signal into an electric signal, and examples thereof include a photodiode, an avalanche photodiode, a phototransistor, a photomultiplier tube (photomultiplier) using a photoelectric effect, and a photoconductive element using a change in electric resistance due to light irradiation.

(42) An output of the light sensor **302** is converted into a digital detection signal by an A/D converter and supplied to the processing device **500**. The detection signal indicates a time waveform I.sub.OBJ(t) of the object light S.sub.OBJ.

(43) The processing device **500** generates a spectrum I.sub.OBJ(λ) of the object light S.sub.OBJ based on an output signal of the light receiving device **300**. Then, transmittance T(λ) of the object OBJ is calculated based on the spectrum I.sub.IN(λ) of the measurement light S.sub.IN and the spectrum I.sub.OBJ(λ) of the object light S.sub.OBJ.

(44) $T(\ ) = I_{\mathrm{OBJ}}(\ ) / I_{IN}(\ )$ (2)

(45) A part of the measurement light S.sub.IN may be split into another path using a beam splitter or the like on the lighting device **200** side of the object OBJ, and the time waveform I.sub.IN(t) of the split measurement light S.sub.IN may be measured by a light receiving device (not illustrated in FIG. **2**) different from the light receiving device **300** to obtain the spectrum I.sub.IN(λ) of the measurement light S.sub.IN. Alternatively, when the stability of the measurement light S.sub.IN is high, the spectrum I.sub.IN(λ) measured in advance can be held and used.

(46) FIG. **4** is a graph for explaining spectroscopy by the optical measurement device **100** in FIG. **2**. As described above, the time t and the wavelength λ of the measurement light S.sub.IN correspond to each other on a one-to-one basis, and thus the waveform I.sub.IN(t) of a time domain of the measurement light S.sub.IN can be converted into the spectrum I.sub.IN(λ) of a frequency domain.

(47) The time waveform I.sub.OBJ(t) of the object light S.sub.OBJ generated from the measurement light S.sub.IN also has the time t and the wavelength λ corresponding to each other on a one-to-one basis. Therefore, the processing device **500** can convert the waveform I.sub.OBJ(t) of the object light S.sub.OBJ indicated by an output of the light receiving device **300** into the spectrum I.sub.OBJ(λ) of the object light S.sub.OBJ.

(48) The processing device **500** can calculate a transmission spectrum T(λ) of the object OBJ based on a ratio I.sub.OBJ(λ)/I.sub.IN(λ) of the two spectra I.sub.OBJ(λ) and I.sub.IN(λ).

(49) It is assumed that the relationship between the wavelength λ and the time t in the measurement light S.sub.IN is expressed by a function λ=f(t). Most simply, the wavelength λ linearly changes with respect to the time t according to a linear function. When the time waveform I.sub.OBJ(t) of the object light S.sub.OBJ decreases at a certain time t.sub.x, it means that the transmission spectrum T(λ) has an absorption spectrum at a wavelength λ.sub.x=f(t.sub.x).

(50) Note that the processing in the processing device **500** is not limited thereto. The transmission spectrum T(λ) may be calculated by calculating a ratio T(t)=I.sub.OBJ(t)/I.sub.IN(t) of two time waveforms I.sub.OBJ(t) and I.sub.IN(t) of time and then converting a variable t of the time waveform T(t) into λ.

(51) The above is the configuration of the optical measurement device **100**. According to the optical measurement device **100**, the object OBJ is irradiated with the measurement light S.sub.IN from the support portion **401** side, and the diffusely transmitted light emitted from a surface of the object OBJ opposite to the support portion **401** is detected as the object light S.sub.OBJ.

(52) As a result, the object light S.sub.OBJ is not shielded by the support portion **401**, and thus more object light S.sub.OBJ can be taken in by the light receiving device **300**, and utilization

efficiency of light can be improved.
(53) Next, a specific example of the optical measurement device **100** will be described.
Example 1
(54) FIG. **5** is a view illustrating an optical measurement device **100**A according to Example 1. The conveyance device **400** includes a plurality of the support portions **401**, and each of the support portions **401** is configured to be able to support (fix) the object OBJ. A cross-sectional view of the support portion **401** is illustrated in a lower part in FIG. **5**. The support portion **401** has a recessed portion **410** into which the object OBJ fits. The recessed portion **410** is formed on a common surface **406** and has a larger diameter than the object OBJ. In addition, a through hole **412** having a smaller diameter than the object OBJ is formed in the bottom surface of the recessed portion **410**. The through hole **412** corresponds to the above-described opening **402**. The object OBJ is placed to straddle the through hole **412**. The measurement light S.sub.IN passes through the through hole **412** and the bottom surface side of the object OBJ is radiated with the measurement light S.sub.IN. The light receiving device **300** is provided on an upper side of the irradiation region **10** so as to face the upper surface of the object OBJ.
(55) The conveyance device **400** further includes a suction box **430**. A negative pressure is maintained inside the suction box **430** by an exhaust pump (not illustrated), the through hole **412** serves as an intake port, and the object OBJ is sucked to the through hole **412**. With this configuration, the object OBJ can be prevented from falling off from the recessed portion **410**.
(56) The lighting device **200** includes a folding mirror **202**. The folding mirror **202** is provided inside the conveyance device **400**, more specifically, inside the suction box **430** and immediately below the through hole **412** that is the opening **402**. A light source head **204** that generates the measurement light S.sub.IN is provided outside the suction box **430**, and a window **432** through which the measurement light S.sub.IN passes is provided on a side surface of the suction box **430**. The measurement light S.sub.IN is incident on the folding mirror **202** from the side surface of the conveyance device **400** through the window **432**. The folding mirror **202** reflects the measurement light S.sub.IN toward the object OBJ. The size of a window **434** can be reduced as long as the measurement light S.sub.IN can pass through the window, and the influence on the pressure in the suction box **430** is sufficiently small. The window **434** may be a glass transparent to a wavelength band of the measurement light S.sub.IN, and in this case, the influence on the pressure of the window **434** can be eliminated.
Example 2
(57) FIG. **6** is a view illustrating an optical measurement device **100**B according to Example 2.
(58) In Example 2, the bottom surface of the object OBJ is directly irradiated with the measurement light S.sub.IN from the lower side of the conveyance device **400** without using the folding mirror. For example, a window **434** is provided on a bottom surface of the suction box **430**, the measurement light S.sub.IN can pass through the window **434** and the through hole **412**, and the bottom surface of the object OBJ can be irradiated.
(59) In Example 2, the folding mirror is not required as compared with Example 1, and thus the configuration can be simplified.
(60) In Example 1 and Example 2, when the object OBJ is not sucked, the suction box **430** can be omitted.
Example 3
(61) FIG. **7** is a view illustrating an optical measurement device **100**C according to Example 3. In Example 3, vacuum suction by the suction box **430** is used to support the object OBJ against gravity in the recessed portion **410**. The light receiving device **300** is disposed on the lower side of the conveyance device **400**. Similar to Example 1, a folding mirror may be disposed inside the suction box **430**, and the measurement light S.sub.IN generated by the light source head **204** may be incident on the folding mirror from the side of the suction box **430**.
Example 4

(62) FIGS. **8**A to **8**C are views each illustrating a configuration example of the support portion **401**. The conveyance device **400** is a movable table, and includes a plurality of holders **450** formed integrally with or separably from a table **440**. The plurality of holders **450** are disposed at equal intervals or at unequal intervals in the conveyance direction.
(63) FIG. **8**B illustrates a cross-sectional view of the support portion **401**. In the holder **450**, a recess **452** having a diameter slightly larger than the object OBJ and an opening **454** having a diameter smaller than the object OBJ are formed. The object OBJ is supported and conveyed while being accommodated in the recess **452** of the holder **450**.
(64) An opening **442** is formed in the table **440** so as to overlap the opening **454**. The opening **442** and the opening **454** correspond to the opening **402** described above.
(65) The object OBJ to be inspected is mounted in the recess **452** of the holder **450** on the upstream side (not illustrated).
(66) As illustrated in FIG. **8**C, a recess **456** may be formed so as to penetrate the holder **450**.
Example 5
(67) FIG. **9** is a view illustrating an optical measurement device **100**E according to Example 5. In Example 5, the conveyance device **400** is a roller conveyor, and includes a plurality of rollers **422** provided apart from each other in the conveyance direction. In this Example 5, a gap between two rollers **422** adjacent to each other can be used as the opening **402** described above.
Example 6
(68) When the lighting device **200** can be sufficiently downsized, the lighting device **200** may be disposed in the suction box **430**.
Example 7
(69) FIG. **10** is a view illustrating an optical measurement device **100**F according to Example 7. In the above description, the case where the opening **402** is provided in the support portion **401** of the conveyance device **400** has been described, but the present invention is not limited thereto. In Example 7, the opening **402** of the support portion **401** is omitted, and instead, the support portion **401** is made of a transparent material such as glass or resin. The lighting device **200** irradiates the first surface of the object OBJ with the measurement light S.sub.IN having the wavelength changing over time through the transparent support portion **401**. The light receiving device **300** detects the diffusely transmitted light S.sub.OBJ emitted from the second surface of the object OBJ.
(70) Advantages of the optical measurement device **100** F will be described. The advantages of the optical measurement device **100** F are clarified by comparison with a comparative technique. FIG. **11** illustrates an optical measurement device **100**R according to the comparative technique. In the comparative technique, the support portion **401** is made of glass or the like, the object OBJ is irradiated with the measurement light S.sub.IN from the upper side, and the object light S.sub.OBJ is measured by the light receiving device **300** on the lower side of the support portion **401**.
(71) The maximum incident intensity I.sub.DET(MAX) of the light incident on the light receiving device **300** R is expressed as
(72) $I_{\mathrm{DET(MAX)}} = I_{\mathrm{MAX}} \times \quad \times$, where α is the transmittance of the object light OBJ, B is the transmittance of the support portion **401**, and I.sub.MAX is the maximum intensity with which the object OBJ can be irradiated, and as the transmittance β of the support portion **401** decreases, the maximum incident intensity I.sub.DET(MAX) decreases, and the measurement accuracy decreases.
(73) On the other hand, in Example 7, the intensity I.sub.IN of the light emitted from the lighting device **200** can be adjusted such that the intensity I.sub.IN′=I.sub.IN×β of measurement light S.sub.IN′ transmitted through the support portion **401** becomes the maximum intensity I.sub.MAX.
(74) $I_{IN} = I_{\mathrm{MAX}} /$
(75) At this time, the maximum incident intensity I.sub.DET(MAX) of the light incident on the light receiving device **300** is expressed as
(76) $I_{\mathrm{DET(MAX)}} = I_{\mathrm{IN}} \times \quad \times \quad = I_{\mathrm{MAX}} \times$, and the light receiving device **300** can detect more light

without being affected by attenuation and light shielding in the support portion **401**, and the measurement accuracy can be improved.

Light Receiving Device

(77) The present inventors have recognized that when an object such as food and drink obtained by solidifying powder in a solid state is to be inspected, the diffusely transmitted light that is the object light S.sub.OBJ has low directivity and is emitted over a wide range. In the case of an object having extremely low transmittance (For example, when a fine powder is hardened to have a thickness of about 3 mm, the transmittance is several % or less, more specifically, 1% or less.), as much object light S.sub.OBJ as possible is required to be detected for highly accurate spectroscopy.

(78) FIG. **12** is a view illustrating a configuration example of the light receiving device **300**. The light receiving device **300** includes the light sensor **302** and a condensing optical system **310**. In the spectroscopy using the chirped pulsed light, high-speed response of the light sensor **302** is required, and the response of the light sensor **302** is more advantageous as the area thereof is smaller. Therefore, as a light receiving portion of the light sensor **302** having sensitivity, a portion smaller than the object OBJ may be used, and for example, a portion having a diameter of about 0.5 to 1 mm may be used. On the other hand, the condensing optical system **310** is provided in order to cause the object light S.sub.OBJ diffused in a wide range from the object OBJ to be incident on the light sensor **302** having the smaller light receiving portion. The condensing optical system **310** can typically use one or a plurality of lenses **312**.

(79) The condensing optical system **310** includes the lens **312** having a diameter sufficiently larger than the object OBJ. By using the lens **312** larger than the object OBJ, the diffusely transmitted light can be condensed on the light sensor **302** as much as possible.

Applications

(80) Next, applications of the optical measurement device **100** according to the embodiment will be described. The optical measurement device **100** can be used for an inspection device for the product such as food and drink obtained by solidifying powder in a solid state. FIG. **13** is a view illustrating an inspection device **800** that is one embodiment of the optical measurement device **100**. The inspection device **800** inspects a large amount of the product P such as food and drink and determines whether the product P is good. In the case of food and drink, the transmittance thereof is on the order of 1/100 to 1/1000.

(81) As described with respect to the optical measurement device **100**, the inspection device **800** includes the lighting device **200**, the light receiving device **300**, the conveyance device **400**, and the processing device **500**. Further, the inspection device **800** includes a light receiving device **810**, a beam damper **820**, a digitizer **830**, and a pump **840**.

(82) The lighting device **200** includes a light source **210**, a pulse stretcher **220**, and an irradiation optical system **230**. The light source **210** generates coherent pulsed light having a broad continuous spectrum in a continuous spectrum of at least 10 nm, specifically in the near infrared region of 900 to 1300 nm. The light source **210** may be a super continuum (SC) light source including a pulsed laser and a nonlinear element. As the pulse laser, a mode-locked laser, a microchip laser, a fiber laser, or the like can be used. As the nonlinear element, a nonlinear fiber such as a photonic crystal fiber can be used.

(83) The pulse stretcher **220** extends the pulse width of the pulsed light generated by the light source **210** in such a manner that the time and the wavelength correspond to each other on a one-to-one basis. The pulse stretcher **220** may include one wavelength dispersion fiber.

(84) Alternatively, the pulse stretcher **220** may include a demultiplexer that splits the pulsed light into a plurality of paths for each wavelength, a plurality of fibers (fiber bundle) that gives a different delay to each of the plurality of paths, and a multiplexer that recombines outputs of the plurality of fibers. The demultiplexer can include planar lightwave circuits (PLC), and may specifically include array waveguide grating (AWG). The plurality of fibers constituting the fiber bundle has different lengths.

(85) The support portion **401** of the conveyance device **400** includes the recessed portion **410**. A plurality of the products P are each placed inside the recessed portion **410** by a mounter (not illustrated) on the upstream side (left side in the drawing). The conveyance device **400** moves the plurality of support portions **401** in the arrangement direction (right direction in the drawing) thereof. Of the surfaces of the recessed portion **410**, a surface on which the product P is placed is referred to as a front surface, and the opposite surface is referred to as a back surface.
(86) The irradiation optical system **230** irradiates the irradiation region **10** with the expanded pulse as the measurement light S.sub.IN. The irradiation region **10** is defined at a passing portion of the product P, that is, a passing portion of the recessed portion **410**. The irradiation optical system **230** can include a transmission optical system such as a lens, a reflection optical system such as a mirror, or a combination thereof. As the recessed portion **410** moves, the plurality of products P sequentially cross the irradiation region **10**.
(87) The light source **210** repeatedly generates pulsed light at a predetermined frequency (cycle). An operating frequency of the light source **210** may be determined according to a moving speed of the recessed portion **410**, that is, the conveying speed of the product P. The operating frequency is determined such that the same product P is irradiated with a plurality of pieces of the measurement light S.sub.IN while one product P is present in the irradiation region **10**.
(88) The operation of the light source **210** is independent of the operation of the conveyance device **400**, in other words, the position of the product P. Therefore, even when the product P is not present in the recessed portion **410**, the irradiation region **10** is repeatedly irradiated with the measurement light S.sub.IN.
(89) The light receiving device **300** is provided on an upper side of the recessed portion **410**. The through hole **412** is formed in the bottom surface of the recessed portion **410**. The through hole **412** is formed in order to guide the measurement light S.sub.IN from the irradiation optical system **230** to the bottom surface of the product P.
(90) The pump **840** may be provided on the back surface side of the recessed portion **410**. The pump **840** constitutes a suction means, and the product P is sucked to the recessed portion **410** by setting the back surface side of the recessed portion **410** to a negative pressure, and the product P can be prevented from falling off from the recessed portion **410** along with the conveyance of the product P.
(91) The light receiving device **300** measures the time waveform I.sub.OBJ(t) of the object light S.sub.OBJ. The beam damper **820** is provided on the optical axis OA**2** of the measurement light S.sub.IN in order to prevent stray light.
(92) The light receiving device **810** is provided in order to measure the spectrum of the measurement light S.sub.IN. The irradiation optical system **230** splits a part of the measurement light S.sub.IN into another arm as reference light S.sub.REF using a beam splitter or the like. The light receiving device **810** measures a time waveform I.sub.REF(t) of the reference light S.sub.REF split into another arm. The time waveform I.sub.REF(t) is equivalent to the time waveform I.sub.IN(t) of the measurement light S.sub.IN.
(93) The digitizer **830** includes an A/D converter, samples outputs of the light receiving device **300** and the light receiving device **810**, that is, time waveforms I.sub.OBJ(t) and I.sub.REF(t) at a predetermined sampling frequency, and converts the sampled time waveforms into waveform data D.sub.OBJ(t) and D.sub.IN(t) of digital signals. In the case of using the light receiving devices **300** and **810** of digital outputs, the digitizer **830** can be omitted.
(94) The processing device **500** processes the waveform data D.sub.OBJ(t) and D.sub.IN(t) of digital to obtain a transmission characteristic (or absorption characteristic) T(λ) of the product P. The processing device **500** may be implemented in a combination of general or special purpose computers including a storage medium such as a processor, a memory, a hard disk, and the like, and a software program. The processing of the processing device **500** is as described above.
(95) According to the inspection device **800**, the product P is irradiated with the measurement light

S.sub.IN from the conveyance device **400** side, and thus the object light S.sub.OBJ can be prevented from being shielded by the through hole **412**, and more object light S.sub.OBJ can be detected by the light receiving device **300**.

(96) Note that in the inspection device **800** in FIG. **13**, when the product P is not present in the recessed portion **410**, the measurement light S.sub.IN passes through the through hole **412** and leaks to the light receiving device **300** side. Assuming that the light receiving device **300** is disposed on the optical axis OA**2** of the measurement light S.sub.IN, the measurement light S.sub.IN having a high intensity is directly incident on the light sensor **302** when the product P is not present on the optical axis OA**2**, which is not preferable. Therefore, the light receiving device **300** may be configured such that the measurement light S.sub.IN is not incident on the light sensor **302** when the product P is not present in the recessed portion **410**.

(97) For example, the light receiving device **300** is configured such that a component S.sub.e of the measurement light S.sub.IN of the diffusely transmitted light (object light S.sub.OBJ) of the object OBJ emitted in a direction (deviation angle is θ) deviated from the optical axis OA**2** is incident on the light sensor **302**.

(98) Note that the object light S.sub.OBJ may be incident on an incident aperture of the light receiving device **300**, as long as the object light S.sub.OBJ in the direction of the optical axis OA**2** is not incident on the light sensor **302**.

(99) Thus, the light receiving device **300** can be protected when the product P is not present in the recessed portion **410**. At this time, the light source **210** of the lighting device **200** can be freely running asynchronously with the operation of the conveyance device **400**, and shutter control synchronized with the operation of the conveyance device **400** is also not required.

(100) The embodiments merely illustrate the principle and applications of the present invention, and many modifications and changes in arrangement can be made in the embodiments without departing from the spirit of the present invention defined in the claims.

Claims

1. An optical measurement device comprising: a conveyance device configured to support and convey an object and including a support portion configured to support the object; a lighting device configured to irradiate a first surface of the object with measurement light having a wavelength changing over time through the support portion, wherein the measurement light is coherent laser light; and a light receiving device configured to detect diffusely transmitted light emitted from a second surface of the object, wherein the support portion has an opening narrower than the object, the lighting device irradiates the first surface of the object with the measurement light through the opening of the support portion, a negative pressure is maintained inside the conveyance device, and the opening serves as an intake port such that the object is sucked to the conveyance device.
2. The optical measurement device according to claim 1, wherein the conveyance device includes a plurality of the support portions each configured to support the object, each of the plurality of support portions including a recessed portion provided on a common surface of the conveyance device, and the opening is a through hole formed in a bottom surface of the recessed portion.
3. The optical measurement device according to claim 1, wherein the lighting device includes a light source structured to generate the measurement light, the conveyance device has a window through which the measurement light is introduced into the conveyance device, and the lighting device includes a folding mirror provided inside the conveyance device, the folding mirror reflects the measurement light which has propagated through the window toward the first surface of the object.
4. The optical measurement device according to claim 1, wherein the light receiving device includes: a light sensor having a size smaller than the object; and a condensing optical system including a lens having a size larger than the object.

5. The optical measurement device according to claim 1, wherein the light receiving device includes a light sensor, and a component of the diffusely transmitted light of the object emitted in a direction deviated from an optical axis of the measurement light is incident on the light sensor.
6. An optical measurement device comprising: a conveyance device configured to support and convey an object, and including a support portion configured to support the object; a lighting device configured to irradiate a first surface of the object with measurement light having a wavelength changing over time, wherein the measurement light is coherent laser light; and a light receiving device configured to detect diffusely transmitted light emitted from a second surface of the object, wherein the light receiving device includes: a light sensor having a size smaller than the object; and a condensing optical system including a lens having a size larger than the object, and the support portion has an opening narrower than the object, the lighting device irradiates the first surface of the object with the measurement light through the opening of the support portion, a negative pressure is maintained inside the conveyance device, and the opening serves as an intake port such that the object is sucked to the conveyance device.